Volatility Risk Premium and Its Capture

What does this all mean to a trader or technical analyst? The positive spread between implied volatility and realized volatility offers an opportunity to capture this differential by selling options. A positive spread is when the implied volatility is greater than the realized volatility.

The volatility risk premium (VRP) represents the compensation that investors earn for providing protection against unexpected market volatility. By selling options contracts, investors are essentially underwriting insurance; as with all insurance, the underwriter seeks a risk premium.

In the case of VIX, levels below 20 could consistently result in a capture of this VRP. This can be achieved through S&P 500® options at the index level or at the individual stock level. The VRP exists in all asset classes because market participants want protection for various reasons. The buyer of the option typically wants to hedge against losses. The seller of the option, on the other hand, wants to profit from the selling or underwriting of the option. The option buyer pays a premium, an upfront cost, which is collected by the option seller. Collecting this option premium is capturing the VRP.

As an example, a trader who is long the S&P 500 (or a portfolio with a high correlation) can choose to sell S&P 500 call options to capture the existing premium between the implied and realized volatility when the implied volatility is greater than realized volatility. Other possibilities include selling call or put options or options spreads against a basket of U.S. Treasuries to capture this premium, thereby enhancing returns.

Performance of a cash-covered put strategy (www.cboe.com/PUT) and a covered call strategy (www.cboe.com/BXM) are compared with a "buy and hold" S&P 500 strategy in Figure 6.2.2. Details of these strategies are also available at the web addresses above.

During the period June 30, 1986 to June 30, 2022, both strategies had a lower drawdown than the S&P 500 Index (-32.7 and -35.8 vs. -51%) with lower returns (9.4 and 8.2 vs. 10.2). They also had lower standard deviations than the S&P 500 (10.2 and 10.9 vs. 15.3) and Betas of 0.57 and 0.63 to the S&P 500. Given these metrics, they could serve as useful diversification to a portfolio that utilizes a "buy and hold" method on the S&P 500.

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6.2 Volatility Risk Premium

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1986 - Jun 30, 2022)

Index	Maximum drawdown	Annualized return	Standard deviation	Beta vs market	Sha rat
PUT	−32.7	9.4	10.2	0.57	0.(
BXM	−35.8	8.2	10.9	0.63	0.4
S&P 500 Index	−51	10.2	15.3	1	0.4

FIGURE 6.2.2 PUT and BXM strategies versus SPX buy and hold
Source: www.cboe.com

In summary:

- When implied volatility is less than realized volatility, one might want to purchase volatility with long calls or call spreads or long puts or put spreads to capture this differential.
- When implied volatility is greater than realized volatility, one might want to sell volatility.

This is one way to express a view of market direction without committing to the underlying asset. In addition to S&P 500 options, VIX options and futures are also available for application to a variety of tail risk hedging, portfolio risk management, and diversification as well as alpha generation strategies.[1]

The principle of VRP capture applies equally to individual stocks and all optionable asset classes, where the implied volatility is higher or lower than its realized volatility on average, providing traders an incentive to position trades using individual options strategies described above.

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[1] Moran, Matthew and Liu, Berlinda. "The VIX Index and Volatility-Based Global Indexes and Trading Instruments - A Guide to Investment and Trading Features." *CFA Institute Research Foundation Briefs*, April 2020. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3668983.

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The Differential Between Implied and Realized Volatility

An analysis of implied volatility calculated from VIX and the subsequent realized volatility, between 2004 and 2022, is shown in Figure 6.2.1. VIX values are shown in green and the subsequent 21-day realized volatility in dark blue. During this period, when VIX was at 20 or below, implied volatility overestimated realized volatility. The spread is observed to narrow when VIX values were greater than 20, particularly during VIX spikes over 40. In some instances, VIX estimates of forward volatility dropped below the subsequent realized volatility. Notable instances include VIX spikes in 2008 (Great Financial Crisis) and 2020 (COVID-19).

Statistically speaking, VIX overestimated realized volatility by 3.42 points on average with a median value of 3.93 points during this period, as shown in the table enclosed in the chart in Figure 6.2.1.

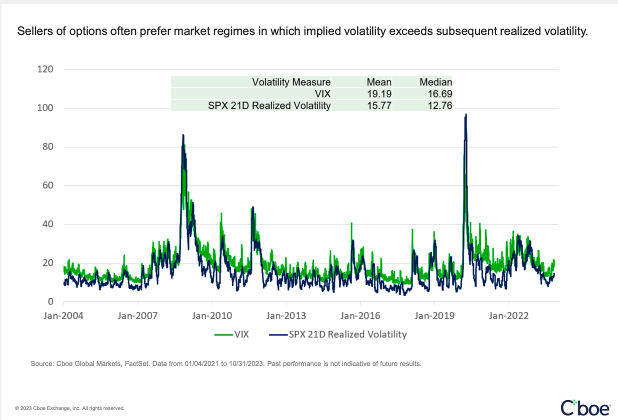

FIGURE 6.2.1 Implied (VIX, green line) and 21-day realized volatility (blue line), January 2004-January 2022

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